

Empirical Research in Finance

2. Models and data

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From Research Question to Data

Once you have a research question. . .

- ▶ You need to decide **what type of data** can answer it.
- ▶ Consider **what analysis methods** are appropriate.
- ▶ Always ask: **can I establish correlation or causation?**

Example:

- ▶ Research Question: “Does ESG disclosure reduce cost of debt of firms?”
- ▶ Required data: firm-level debt costs, ESG scores, control variables over time $\implies$ panel data.

The Main Types of Data in Finance

1. **Cross-sectional:** different entities, one point in time.
Example: returns for all S&P500 firms on Jan 1, 2025.
2. **Time-series:** one entity, many time periods.
Example: daily Apple stock prices from 2010–2025.
3. **Panel:** multiple entities over time.
Example: stock returns of all S&P500 firms from 2010–2025.

Cross-sectional data: x_i , where $i = 1, \dots, N$

Time-series data: x_t , where $t = 1, \dots, T$

Panel data: x_{it} , where $i = 1, \dots, N$
and $t = 1, \dots, T$

Why Data Type Matters

- ▶ Determines your methodology: OLS regression, Fama-French model, event study, etc.
- ▶ Limits what you can conclude: correlation or causation?

Correlation vs. Causation

- ▶ Correlation: two variables move together.
- ▶ Causation: one variable actually changes the other.

Examples:

- ▶ “Countries with more Starbucks have higher GDP.”
→ richer countries may simply have more demand for Starbucks.
- ▶ “Bitcoin rises when Elon Musk tweets.”
→ both may be driven by broader investor attention or market hype.

Non-Random Assignment: Why Causality Is Unclear

- ▶ A bank has 10 branches.
- ▶ It gives new trading software to 5 of them.
- ▶ If assignment is *not* random, there is a temptation to choose branches strategically.
 - ▶ E.g. Suppose it gives the software to its best branches first.
 - ▶ Better branches may have better managers, wealthier clients, better locations
- ▶ If those branches earn more, why is that?
 - ▶ Because of the software – or because they were already better branches?
- ▶ Without **random assignment**, we cannot tell.

Profit difference = software effect + branch differences

How Random Assignment Helps Identify Causality

- ▶ Instead, assign the software randomly.
- ▶ 5 branches get the new software, 5 do not.
- ▶ Then compare profits after 1 month.
- ▶ Random assignment breaks the link between **treatment** and **branch characteristics**.
- ▶ So profit differences can be interpreted causally.

Profit difference $\approx$ software effect

Causation and correlation

- ▶ Natural or controlled experiments are typically not feasible in finance
- ▶ This makes it hard, and sometimes impossible, to infer causality. But we can always talk about **correlations**.
- ▶ Correlations can still be useful to predict outcomes and identify patterns that deserve further investigation.
 - ▶ E.g.: Stock prices are correlated with consumer confidence. Even if not causal, it is still useful to describe the state of the economy and improve predictions.

Ordinary Least Squares

- ▶ Economic model: $y = f(x)$
 - ▶ Take a linear approximation: $f(x) = \beta_0 + \beta_1 x$
 - ▶ Acknowledge that there are things that affect y outside the economic model.

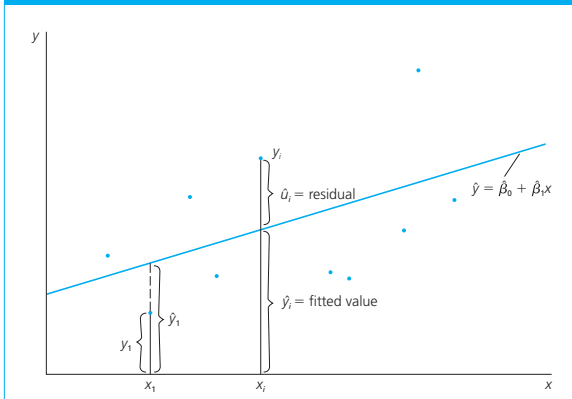
⇒ Cross-sectional, simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

where $E[u_i] = 0$.

- ▶ The *Ordinary Least Squares* (OLS) procedure
 - ▶ picks estimates $\hat{\beta}_0$ and $\hat{\beta}_1$
 - ▶ to minimize the sum of squared residuals $\hat{u}_i$.
- ▶ Under very few assumption, OLS works.
Simple and robust ⇒ usually the right choice.

FIGURE 2.4 Fitted values and residuals.



Source: Wooldridge

Terminology: *run the regression of y on x or regress y on x*

Reading:

- ▶ Wooldridge, chapter 1 (causality), section 2.1 & 2.2